

**Course-** BTech  
**Course Code-** CSET101

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**Date-**

**Type-** Core  
**Course Name-** Computational  
Thinking and Programming  
**Semester-** odd  
**Batch-** ALL

### Tutorial Assignment: 3 Solution

1) a

2) d

3. b

**Explanation-** (Python interpreter will show invalid syntax, as if you want to do simultaneous assignment to variable, so they should be separated by comma (,)).

4. a

**Explanation-** printf is not defined in python. (Python interpreter will show NameError: name 'printf' is not defined for option b and c)

**5. Output Error - ( NameError: name 'User\_name' is not defined)**

**Explanation-** Since python is case sensitive language so user\_name and User\_name both are treated as different.

**6. Output -: 14**

**Explanation-** | is bitwise or operator so it would perform or operation between 1010 | 1100 and result would be 1110 which is basically 14 in decimal.

**7. Output:- 1**

**Explanation-** & is bitwise and operator so it would perform and operation between 0101 & 1001 and result would be 0001 which is basically 1 in decimal.

### 8. Output is -

`var = 2`

`var = 4`

`var = 8`

`var = 16`

`var = 32`

`var = 64`

#### Explanation-

The code uses the left shift operator (`<<`) to shift the binary representation of the variable `var` by a specified number of positions to the left.

Here's how the calculations work:

- `var << 0` shifts `var` by 0 positions to the left, which has no effect, so it remains 2.
- `var << 1` shifts `var` by 1 position to the left, effectively multiplying it by 2, so it becomes 4.
- `var << 2` shifts `var` by 2 positions to the left, effectively multiplying it by 4, so it becomes 8.
- `var << 3` shifts `var` by 3 positions to the left, effectively multiplying it by 8, so it becomes 16.
- `var << 4` shifts `var` by 4 positions to the left, effectively multiplying it by 16, so it becomes 32.
- `var << 5` shifts `var` by 5 positions to the left, effectively multiplying it by 32, so it becomes 64.

### 9)

```
import math
```

```
startTimen = (5*60 + 43)
```

```
slowTime1 = 4 * (9*60 + 32)
```

```
fastTime = 3*(7*60+12)
```

```
slowTime2 = 2.5*(9*60+32)
```

---

```
totalTime = (slowTime1+fastTime+slowTime2) # in seconds
```

```
totalTimeInMins = math.ceil(totalTime/60)
```

```
timeWhenReach = startTimen + totalTimeInMins #in mins
```

```
print("Sam will reach home at = " ,format(timeWhenReach/60,".2f"))
```

**10)**

```
# Function to calculate the restaurant tab
```

```
def calculate_tab():
```

```
    # Get input from person 1
```

```
    person1_item = input("Enter item for person 1: ")
```

```
    person1_price = float(input(f"Enter the price of {person1_item}: "))
```

```
    # Get input from person 2
```

```
    person2_item = input("Enter item for person 2: ")
```

```
    person2_price = float(input(f"Enter the price of {person2_item}: "))
```

```
    # Get the gift certificate amount
```

```
    gift_certificate = float(input("Enter the gift certificate amount: "))
```

```
    # Calculate the total cost
```

```
    total_cost = person1_price + person2_price
```

```
    # Calculate the restaurant tax (8%)
```

```
    tax_rate = 0.08
```

```
    restaurant_tax = total_cost * tax_rate
```

---

```
# Calculate the remaining amount on the gift certificate
```

```
remaining_amount = gift_certificate - total_cost - restaurant_tax
```

```
# Display the results
```

```
print(f"Total cost: ${total_cost:.2f}")
```

```
print(f"Restaurant tax (8%): ${restaurant_tax:.2f}")
```

```
if remaining_amount >= 0:
```

```
    print(f"Remaining amount on the gift certificate: ${remaining_amount:.2f}")
```

```
else:
```

```
    print(f"You owe an additional ${abs(remaining_amount):.2f}.")
```

```
# Call the function to calculate the tab
```

```
calculate_tab()
```